

Capstone Project Proposal, Outline, and Proposal Draft Versions by All 3 Teammates

Below is the Project Proposal and Proposal Outline information from Canvas, followed by sections for all teammates' draft versions of Project Proposals for our capstone project, for the purpose of collaborative teamwork and combining our final proposal as a blend of all teammates' versions.

Project Proposal:

Objective:

- Submit a formal proposal for your Capstone Project based on the provided Capstone Project Outline. This proposal will help you clarify your project goals, scope, and timeline before development begins.

Proposal Requirements:

- Your proposal must include the following sections:
 - Project Title and Team Members
 - Title of your project
 - Names and roles of all team members
 - Project Description
 - A clear and concise summary of your project idea
 - The problem your project aims to solve
 - Target users and use cases
 - Technology Stack
 - Languages, frameworks, libraries, and tools you plan to use
 - Justification for your choices
 - Features and Functionality
 - List of core features
 - Any stretch goals or optional features
 - Milestones and Timeline
 - Breakdown of tasks and estimated completion dates
 - Weekly goals aligned with the course schedule
 - Challenges and Risks
 - Potential technical or logistical challenges
 - Mitigation strategies
 - Evaluation Criteria
 - How you plan to measure success
 - What a “complete” project looks like

Tips:

- Be realistic and specific.
- Use diagrams or mockups if helpful.
- This proposal will guide your development and be used for feedback.

Capstone Project Outline:

- Project Scope:
 - Goal: Build a platform that ingests business data, applies AI for predictions/recommendations, and presents insights via a web and mobile app.
- Tracks:
 - Artificial Intelligence (AI): Predictive modeling, Natural Language Processing (NLP), or recommendation engine.
 - Business & Data Analytics (BDA): Data cleaning, Exploratory Data Analysis (EDA), dashboards, Key Performance Indicators (KPIs) definitions.
 - Software Development (SD): Full-stack app, API integration, deployment.

Week	Milestone	Activity
1	Kickoff & Planning	Define problem, roles, tech stack, success metrics. Gather datasets.
2	Design Phase	Data schema, model architecture, UI wireframes.
3	Data Prep & EDA	The BDA team cleans data, performs EDA, defines KPIs. The AI team explores features.
4	Model Development	The AI team builds a baseline ML model. BDA finalizes dashboards. SD sets up the backend skeleton.
5	Model Refinement and API Design	AI improves accuracy. SD creates API endpoints for model inference.
6	Front-End Development	SD builds UI components. Integrate dashboards (iframe or API).
7	Integration Sprint	Connect AI model → API → Front-end/Mobile. Test data flow end-to-end.

8	Testing & QA	Unit tests, integration tests, performance checks.
9	Deployment	Deploy app to cloud (AWS/Azure/Heroku). Validate CI/CD pipeline.
10	Final Presentation	Prepare the demo, documentation, and business report.

- Deliverables by Track:
 - AI: Trained model, evaluation report, inference API.
 - BDA: Clean dataset, EDA report, interactive dashboard.
 - SD: Full-stack app, integrated APIs, deployed solution.
- Tools:
 - AI: Python, TensorFlow, Jupyter.
 - BDA: SQL, Pandas, Power BI/Tableau.
 - SD: React/React Native + Node.js/.Net Core, GitHub Actions.

Project Proposal - Draft Version by Elias

1. Project Title and Team Members:

- Title of your project:
 - To be determined
- Names and roles of all team members
 - Leena:
 - Team Lead (is that the wording Milton used? Please correct me if I'm wrong.)
 - Business & Data Analytics (BDA): Data cleaning, Exploratory Data Analysis (EDA), dashboards, Key Performance Indicators (KPIs) definitions.
 - Eli:
 - Software Development (SD): Full-stack app, API integration, deployment.
 - Landin:
 - Software Development (SD): Full-stack app, API integration, deployment.

2. Project Description:

- A clear and concise summary of your project idea:
 - This project explores how music spreads across regions and identifies gaps where popular music in one region has not yet reached another. The app will allow users to compare countries, discover hidden artists, and visualize overlap between regional music preferences.
- The problem your project aims to solve:
 - The problem this project aims to solve is that listeners are often not exposed to music they would likely enjoy simply because it hasn't crossed into their region yet. Even though music is globally accessible, discovery is still regionally limited. This project helps users identify globally popular music that is locally unknown.
- Target users and use cases:
 - Target users include music listeners interested in discovery, students exploring globalization and cultural diffusion, and users curious about how music spreads across regions.
 - Use cases include:
 - Comparing music overlap between countries
 - Finding "hidden" artists popular elsewhere but not locally
 - Identifying globally shared vs region-specific music

- Tracking how music spreads geographically over time

3. Technology Stack:

- Languages, frameworks, libraries, and tools you plan to use
 - Frontend:
 - React Native (mobile + web compatibility)
 - React Native Web (optional web support)
 - Maps / Visualization:
 - To be determined
 - Backend:
 - Node.js (lightweight API layer)
 - Express (simple endpoints for data aggregation)
 - Data APIs:
 - To be determined
- Justification for your choices
 - For the already determined choices:
 - React Native allows one UI codebase for mobile and web.
 - Node/Express keeps backend simple for data aggregation.

4. Features and Functionality:

- List of core features
 - Country comparison tool (select 2–3 regions)
 - Overlap percentage between regions
 - “Hidden Music” discovery view
 - Regional popularity lists
 - Map-based exploration (click country → view data)
 - Artist discovery outside user region
 - Global vs local music breakdown
 - Trend comparison between regions
- Any stretch goals or optional features
 - Timeline (where songs appear first)
 - Audio feature/mood comparison (energy, tempo, etc.)
 - Multi-platform popularity comparison

5. Milestones and Timeline:

- Breakdown of tasks and estimated completion dates
- Weekly goals aligned with the course schedule

Week	Milestone	Activity
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1	Kickoff & Planning	• Define problem, roles, tech stack, success metrics. Gather and select datasets and API's.
2	Design Phase	• Define data schema (country, artist, track, platform, date) • Define comparison logic (overlap %, hidden songs, discovery gap) • Create UI wireframes (map screen, comparison screen, hidden music screen) Decide navigation structure and screen flow
3	Data Prep & EDA	The BDA team cleans data, performs EDA, defines KPIs. Utilize API's.
4	Model Development	The AI team builds a baseline ML model. BDA finalizes dashboards (we need clarification on this). SD sets up the backend skeleton: set up backend structure: create API routes, data format, and placeholder endpoints.
5	Model Refinement and API Design	AI improves accuracy. SD creates API endpoints in preparation to plug them into the app screens/menus/lists.
6	Front-End Development	SD builds UI components like screens, menus, lists, etc. Add accessibility features if possible. Integrate dashboards (iframe or API).
7	Integration Sprint	Connect AI model Connect chosen/created APIs → backend → map and comparison UI. Test overlap and hidden-music discovery flow end-to-end.
8	Testing & QA	Test region comparison accuracy, API responses, map interaction, and hidden-music discovery performance.
9	Deployment	Deploy backend and APIs to the cloud. Test production map, region comparison, and discovery features.
10	Final Presentation	Prepare a demo showing map, region comparison, and discovery gap. Finalize documentation and presentation.

6. Challenges and Risks:

- Potential technical or logistical challenges:
 - Possibly some APIs may return the same data with slightly different fields like artist vs artist-name vs artist name, so we may need to account for that and include those in API calls.
 - Potential limits from APIs: I made sure to link free ones/ones that are free up until an extremely high limit but I could have made a mistake and linked one with more friction involved.
 - Map performance on mobile (but they have specific mobile documentation to help this)
 - Large dataset size, so we need to be specific on which data fields we're using
 - Uploading to cloud, I've never done that before

7. Evaluation Criteria:

- How you plan to measure success:
 - Functional criteria:
 - Users can select countries
 - Users can discover artists not in their region
 - Map interaction works correctly
 - Data loads from real APIs
 - Usability criteria:
 - UI is easy to understand
 - Discovery feature is intuitive
 - Country comparison is clear
- What a “complete” project looks like
 - A “complete” project looks like:
 - Working UI (mobile + web)
 - Country comparison feature
 - Hidden music discovery feature
 - Real API data integration
 - Interactive map or region selector
 - Functional backend logic
 - Final demo and documentation

Project Proposal - Draft Version by Leena

1. Project Title and Team Members:

- Title of your project
- Names and roles of all team members
 - Leena- Team lead, BDA
 - Responsibilities: Data gathering, cleaning, storage KPI, EDA, data visualization. Timeline management, manages 'Main' branch?
 - Eli- Dev 1
 - Responsibilities: Front + back ends
 - Landin- Dev 2
 - Responsibilities: Front + back ends

2. Project Description:

- A clear and concise summary of your project idea
 - This project analyzes how popular music overlaps across regions and how those patterns change over time, using chart data to identify shared and region-specific trends.
- The problem your project aims to solve
 - There is no clear way to visualize or compare music popularity across regions, making it difficult to identify which songs and artists are globally shared versus region-specific.
- Target users and use cases
 - Users:
 - Music listeners interested in discovering global trends
 - Users interested in cross-regional comparisons
 - Data-curious users exploring cultural differences in music
 - Use cases:
 - Compare music overlap between two countries
 - Identify songs popular in other regions but not locally
 - Track how global music overlap changes over time
 - Explore regional differences through interactive maps and dashboards

3. Technology Stack:

- Languages, frameworks, libraries, and tools you plan to use
 - Frontend:
 - React Native (mobile + web compatibility)
 - React Native Web (optional web support)

- Maps / Visualization:
 - PowerBI for dashboards
 - Map API
 - React libraries for integrated charts/dashboards
- Backend:
 - .NET8
- Data SOURCES:
 - Kaggle:
 - <https://www.kaggle.com/datasets/dhruvildave/spotify-charts>
 - <https://www.kaggle.com/datasets/asaniczka/top-spotify-songs-in-73-countries-daily-updated?resource=download>
- Data FRAMEWORK:
 - MSSQL, scripts to build locally
- Justification for your choices
 - Previous familiarity with .NET
 - MSSQL works seamlessly with PowerBI, good for large data handling
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4. Features and Functionality:

- List of core features
 - **Hidden Songs Explorer:** Users select a country and view songs that are popular in other regions but not present locally, helping identify region-specific content.
 - **Country Comparison Tool:** Users select two countries to compare:
 - shared songs (overlap count and %)
 - unique songs per country
 - lists of shared and unique songs
 - **Global Overlap Trend:** Visualizes how music overlap between regions changes over time using line charts and summary metrics.
 - **Country Profile View:** Provides a snapshot of a selected country's music:
 - shared vs unique song distribution
 - top shared songs
 - top region-specific songs
 - **Global Distribution Overview:** Shows how many songs are globally shared versus region-specific across all selected countries.
 - **Interactive Globe Visualization:** An interactive globe allowing users to:
 - select countries
 - view relationships between regions
 - explore overlap through visual connections and highlights

- Any stretch goals or optional features

5. Milestones and Timeline:

- Breakdown of tasks and estimated completion dates
- Weekly goals aligned with the course schedule

Week	Milestone	Activity
1	Kickoff & Planning	• Define problem, roles, tech stack, success metrics. Gather and select datasets and API's.
2	Design Phase 4/5-4/11	• Define data schema (country, artist, track, platform, date) • Define comparison logic (overlap %, hidden songs, discovery gap) • Create UI wireframes (map screen, comparison screen, hidden music screen) Decide navigation structure and screen flow
3	Data Prep & EDA 4/12-4/18 LEENA GONE 4/11-4/16	The BDA cleans data, performs EDA, defines KPIs. Database insert + script formation Continue filling out UI framework, skeletons for features, decide what SPs needed
4	Model Development 4/19-4/25	BDA finalizes dashboards. SD sets up the backend skeleton: set up backend structure: create API routes, data format, and placeholder endpoints.
5	Model Refinement and API Design 4/26-5/2	SD creates API endpoints in preparation to plug them into the app screens/menus/lists.
6	Front-End Development 5/3-5/9	SD builds UI components like screens, menus, lists, etc. Add accessibility features if possible. Integrate dashboards (iframe or API).
7	Integration Sprint 5/10-5/16	Connect chosen/created APIs → backend → map and comparison UI. Test overlap and hidden-music discovery flow end-to-end.

8	Testing & QA 5/16-5/23	Test region comparison accuracy, API responses, map interaction, and hidden-music discovery performance.
9	Deployment 5/24-5/30	Deploy backend and APIs to the cloud. Test production map, region comparison, and discovery features.
10	Final Presentation 5/31-6/6	Prepare a demo showing map, region comparison, and discovery gap. Finalize documentation and presentation.

6. Challenges and Risks:

- Potential technical or logistical challenges
 - LARGE amount of data
 - Getting the UI to function as intended
 - Merge conflicts + branches
- Mitigation strategies
 - Thorough concept of data schema, only taking exactly what we need
 - Starting early, dedicating most time to troubleshooting
 - GitHub rules, staying on top of organization

7. Evaluation Criteria:

- How you plan to measure success
 - Are the project requirements met on our scheduled deadlines?
 - Do all features work as intended, and don't break if NOT used as intended?
 - DOES THE PROJECT ANSWER THE PROBLEM OBVIOUSLY?
- What a "complete" project looks like
 - All agreed upon features work as intended, no bugs
 - Looks nice, is intuitive
 - Complete documentation for everything, under a microscope
 - HAS:
 - Data layer, working API integration, working + usable UI
 - Interactive layers
 - Anticipates user behavior and accounts for it (intuitive)

Project Proposal - Draft Version by Landin

1. Project Title and Team Members:

- Title of your project
- Names and roles of all team members
 - Leena-Team lead
 - Tasks: Data gathering, cleaning, storage KPI, EDA, data visualization, Timeline management, manages 'Main' branch
 - Eli-Software Dev 1
 - Tasks- Front End+Back End
 - Landin-Software Dev 2
 - Tasks- Front End+Back End

2. Project Description:

- A clear and concise summary of your project idea
 - Project is used to compare overlap and discover new artist by looking at regional music popularity in different regions
- The problem your project aims to solve
 - Some artist are limited by cultural or regional places this app will help discover small artist or hidden gems that are not as popular in your region but popular in others
- Target users and use cases
 - People who want to discover new artist, People interested in music spread over regions, students who want to discover music and other regional spread
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3. Technology Stack:

- Technology Stack:
- Languages, frameworks, libraries, and tools you plan to use
 - Frontend:
 - React Native (mobile + web compatibility)
 - React Native Web (optional web support)

- Maps / Visualization:
 - To be determined
- Backend:
 - Node.js (lightweight API layer)
 - Express (simple endpoints for data aggregation)
- Data APIs:
 - To be determined
- Justification for your choices
 - For the already determined choices:
 - React Native allows one UI codebase for mobile and web.
 - Node/Express keeps the backend simple for data aggregation.
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4. Features and Functionality:

- List of core features
 - Regional comparison (selection of few regions)
 - Overlaps between each region
 - “Hidden music” discovery page
 - Popularity based on region
 - Map based discovery
 - Discovery of artist outside of home region
 - Trends for each region
- Any stretch goals or optional features
 - Light/Dark mode
 - Timelines for songs
 - Multi-platform popularity comparison

5. Milestones and Timeline:

- Breakdown of tasks and estimated completion dates
- Weekly goals aligned with the course schedule

Week	Milestone	Activity
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2	Design Phase 4/5-4/11	Define data schema (country, artist, track, platform, date) • Define comparison logic (overlap %, hidden

		songs, discovery gap) • Create UI wireframes (map screen, comparison screen, hidden music screen) Decide navigation structure and screen flow
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8	Testing & QA	Test region comparison accuracy, API responses, map interaction, and hidden-music discovery performance.
9	Deployment	Deploy backend and APIs to the cloud. Test production map, region comparison, and discovery features.
10	Final Presentation	Prepare a demo showing map, region comparison, and discovery gap. Finalize documentation and presentation.

6. Challenges and Risks:

- Potential technical or logistical challenges

- Because we have so many API's some will return the same information or slightly altered will have to change things up so this works properly
- The map discovery page performance on mobile
- Large dataset size (could take long time to load)
- Uploading to cloud (we've never done this)

7. Evaluation Criteria:

- How you plan to measure success
 - App is base functional
 - Users can select regions and see the data between them
 - Users can discover Music and other things in other regions
 - The map based discovery works as it should
 - Get data from actual API's
- What a "complete" project looks like
 - Working UI for mobile and web
 - Regional comparison feature
 - Hidden music discovery
 - API integration
 - Interactive map discovery
 - Functional backend logic
 - Final demo and documentation